# A Sharp Cross-Correlation Lower Bound for Separated Fourier-Sparse Signals

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**Abstract**

Let

$$x(t)=\sum_{j=1}^{\ell}\alpha_j \mathrm{e}^{2\pi \mathrm{i} f_j t}, \qquad y(t)=\sum_{k=1}^{r}\beta_k \mathrm{e}^{2\pi \mathrm{i} f'_k t},$$

where the two frequency sets are separated by distance at least $d$. We prove that, whenever $d \geq C\ell r$, there are such signals for which

$$\left|\langle x,y\rangle_{[-1,1]}\right| \geq c\frac{\ell r}{d}\,\|x\|_{L^2[-1,1]}\,\|y\|_{L^2[-1,1]}\,.$$

The proof first establishes a sharp interaction estimate between two algebraic-polynomial spaces under modulation and then passes to clustered Fourier sums by a confluent finite-difference construction. We also preserve a simpler endpoint-kernel proof, which gives the same conclusion under the stronger assumption $d \gtrsim \ell^2 + r^2$ and is sufficient in the balanced regime $\ell \asymp r$.

## 1 Preliminaries

This section fixes the conventions used throughout the paper and records the standard identities needed in the two proofs.

### 1.1 Fourier-sparse signals and separation

An *$m$-Fourier-sparse signal* is a function $x\colon \mathbb{R}\to\mathbb{C}$ that admits a representation

$$x(t)=\sum_{j=1}^{s}\alpha_j \mathrm{e}^{2\pi \mathrm{i} f_j t}, \qquad 0\leq s\leq m,$$

where the frequencies $f_1,\dots,f_s\in\mathbb{R}$ are distinct and the coefficients $\alpha_1,\dots,\alpha_s\in\mathbb{C}\setminus\{0\}$. Its frequency support is

$$\operatorname{supp}\widehat{x} := \{f_1,\dots,f_s\}.$$

Distinct real-frequency exponentials are linearly independent, so this support does not depend on the chosen representation after zero coefficients and repeated frequencies are removed. For finite sets $A,B\subset\mathbb{R}$, define

$$\operatorname{dist}(A,B) := \inf\{|a-b| : a\in A,\ b\in B\},$$

with the convention $\inf\varnothing=+\infty$. When both sets are nonempty, the infimum is a minimum. Thus the condition $\operatorname{dist}(\operatorname{supp}\widehat{x},\operatorname{supp}\widehat{y})\geq d$ means that every frequency of $x$ is at distance at least $d$ from every frequency of $y$.

For an interval $I \subset \mathbb{R}$ and $u, v \in L^2(I)$, set

$$\langle u, v \rangle_I := \int_I u(t)\overline{v(t)}\, dt, \qquad \|u\|_{L^2(I)} := \langle u, u \rangle_I^{1/2}.$$

When $I = [-1, 1]$, we abbreviate $\|u\|_{L^2[-1,1]}$ to $\|u\|_2$. All unqualified integrals in the polynomial arguments are over $[-1, 1]$.

For nonnegative quantities $X$ and $Y$, the notation $X \lesssim Y$ means $X \le CY$ for an absolute constant $C > 0$; the notation $X \gtrsim Y$ has the analogous meaning. Different occurrences may use different absolute constants.

### 1.2 Polynomial and matrix notation

For $m \ge 1$, let

$$\mathcal{P}_m := \{p \in \mathbb{C}[t] : \deg p < m\},$$

viewed as a subspace of $L^2[-1, 1]$, and let $\Pi_m$ denote the orthogonal projection from $L^2[-1, 1]$ onto $\mathcal{P}_m$. For a finite matrix $A$, write $\|A\|_{\mathrm{op}}$ for its operator norm induced by the Euclidean norm and $\|A\|_{\mathrm{HS}} := (\sum_{j,k} |A_{jk}|^2)^{1/2}$ for its Hilbert–Schmidt norm. We use $\mathbf{1}_E$ for the indicator of an event or condition $E$.

### 1.3 Standard Legendre identities

Let $P_n$ be the Legendre polynomial of degree $n$, normalized by $P_n(1) = 1$.

**Lemma 1.1** (Ultraspherical identities)**.** *Let $C_j^{(\lambda)}$ denote the ultraspherical polynomial of degree $j$ and parameter $\lambda$, with $C_j^{(\lambda)} := 0$ when $j < 0$. The identities used below are the following; $P_j^{(\alpha,\beta)}$ denotes the Jacobi polynomial of degree $j$ with parameters $\alpha$ and $\beta$.*

$$(j + \lambda)C_j^{(\lambda)}(t) = \lambda\left(C_j^{(\lambda+1)}(t) - C_{j-2}^{(\lambda+1)}(t)\right), \tag{DLMF 18.9.7}$$

$$\frac{d}{dt}C_j^{(\lambda)}(t) = 2\lambda C_{j-1}^{(\lambda+1)}(t), \tag{DLMF 18.9.19}$$

$$P_j(t) = C_j^{(1/2)}(t) = P_j^{(0,0)}(t). \tag{DLMF 18.7.9}$$

*See [NIS, Eqs. (18.7.9), (18.9.7), and (18.9.19)].*

The following lemma collects the resulting exact Legendre identities used below.

**Lemma 1.2** (Legendre identities)**.** *The orthogonality normalization is recorded in [NIS, Table 18.3.1]; the symmetry and endpoint values are in [NIS, Table 18.6.1]; and the supremum bound is [NIS, Eq. (18.14.1), with $\alpha = \beta = 0$]. For integers $m, n \ge 0$ and $t \in [-1, 1]$,*

$$\int_{-1}^{1} P_n(t)P_m(t)\, dt = \frac{2}{2n+1}\mathbf{1}_{\{m=n\}}, \tag{1}$$

$$P_n(1) = 1, \qquad P_n(-1) = (-1)^n, \tag{2}$$

$$|P_n(t)| \le 1. \tag{3}$$

*Moreover, for every integer $m \ge 1$,*

$$\sum_{j=0}^{m-1} \frac{2j+1}{2} P_j(t) = \frac{1}{2}\big(P_m'(t) + P_{m-1}'(t)\big). \tag{4}$$

*Proof.* Only (4) requires an explanation beyond the formulas displayed in the lemma. Set $\lambda = 1/2$ in the first identity of Lemma 1.1 and multiply by 2. This gives

$$(2j+1)P_j(t) = C_j^{(3/2)}(t) - C_{j-2}^{(3/2)}(t).$$

Next, set $\lambda = 1/2$ and replace $j$ by $j+1$ and $j-1$ in the second identity of Lemma 1.1. Its third identity gives $P_j = C_j^{(1/2)}$, and hence

$$P'_{j+1}(t) = C_j^{(3/2)}(t), \qquad P'_{j-1}(t) = C_{j-2}^{(3/2)}(t).$$

Substitution yields

$$(2j+1)P_j(t) = P'_{j+1}(t) - P'_{j-1}(t),$$

where the term $P'_{-1}$ is interpreted as zero. Therefore

$$\begin{aligned}\sum_{j=0}^{m-1} \frac{2j+1}{2} P_j(t) &= \frac{1}{2}\sum_{j=0}^{m-1} \big(P'_{j+1}(t) - P'_{j-1}(t)\big) \\ &= \frac{1}{2}\big(P'_m(t) + P'_{m-1}(t)\big),\end{aligned}$$

because all intermediate derivative terms cancel. This is also the endpoint Christoffel–Darboux identity. □

**Lemma 1.3** (Polynomial and Jacobi estimates)**.** *Let $P_s^{(\alpha,\beta)}$ denote the Jacobi polynomial of degree $s$ with parameters $\alpha$ and $\beta$, and define*

$$(a)_0 := 1, \qquad (a)_s := a(a+1)\cdots(a+s-1) \qquad (s \ge 1).$$

*The estimates and identity used below are as follows.*

*First, for $\lambda > -1/2$, define*

$$w_\lambda(t) := (1-t^2)^{\lambda-1/2}, \qquad \|f\|_{w_\lambda} := \left(\int_{-1}^{1} |f(t)|^2\, w_\lambda(t)\, dt\right)^{1/2}.$$

*If $v$ has degree at most $n$, then*

$$\big\|v'\big\|_{w_\lambda} \le c_{n,\lambda}\, \|v\|_{w_\lambda}, \qquad c_{n,\lambda} < \frac{(n+1)(n+2\lambda+1)}{2\sqrt{2\lambda+1}}, \quad \text{(Aleksov–Nikolov–Shadrin, Theorem 1.1)}$$

*where $c_{n,\lambda}$ is the best constant in the first inequality; see [ANS16, Thm. 1.1].*

*Second, for $s \ge 1$,*

$$\frac{d}{dt} P_s^{(\alpha,\beta)}(t) = \frac{s+\alpha+\beta+1}{2} P_{s-1}^{(\alpha+1,\beta+1)}(t). \qquad \text{(DLMF 18.9.15)}$$

*See [NIS, Eq. (18.9.15)].*

*Third, if $\alpha \ge \beta > -1$, $\alpha \ge -1/2$, and $-1 \le t \le 1$, then*

$$\left|P_s^{(\alpha,\beta)}(t)\right| \le P_s^{(\alpha,\beta)}(1) = \frac{(\alpha+1)_s}{s!}. \qquad \text{(DLMF 18.14.1)}$$

*See [NIS, Eq. (18.14.1)].*

*Finally, for fixed $\alpha, \beta > -1/2$, $s \ge 1$, and $0 < \theta < \pi$,*

$$\left|P_s^{(\alpha,\beta)}(\cos\theta)\right| \le C_{\alpha,\beta}\, s^{-1/2} \left(\sin\frac{\theta}{2}\right)^{-\alpha-1/2} \left(\cos\frac{\theta}{2}\right)^{-\beta-1/2}, \qquad \text{(Szegő 7.32.6)}$$

*where $C_{\alpha,\beta}$ depends only on $\alpha$ and $\beta$; see [Sze75, Eq. (7.32.6)].*

**Lemma 1.4** (Polynomial and Legendre derivative bounds)**.** *There is an absolute constant $C > 0$ such that every algebraic polynomial $v$ of degree at most $n$ satisfies*

$$\left\|v'\right\|_{L^2[-1,1]} \le Cn^2 \left\|v\right\|_{L^2[-1,1]}. \tag{5}$$

*Furthermore, for $n \ge 1$ and $0 < \theta < \pi$,*

$$\left|P_n'''(\cos\theta)\right| \le C \min\left\{n^6, n^{5/2}(\sin\theta)^{-7/2}\right\}. \tag{6}$$

*Proof.* At $\lambda = 1/2$, the weight in Lemma 1.3 satisfies $w_{1/2}(t) = 1$. Therefore its first inequality becomes

$$\left\|v'\right\|_{L^2[-1,1]} < \frac{(n+1)(n+2)}{2\sqrt{2}} \left\|v\right\|_{L^2[-1,1]}.$$

This implies (5) after enlarging the absolute constant to cover $n = 0$.

We next derive (6). Since $P_n = P_n^{(0,0)}$, three successive applications of the second identity in Lemma 1.3 give

$$\begin{aligned}
P_n'(t) &= \frac{n+1}{2} P_{n-1}^{(1,1)}(t), \\
P_n''(t) &= \frac{(n+1)(n+2)}{4} P_{n-2}^{(2,2)}(t), \\
P_n'''(t) &= \frac{(n+1)(n+2)(n+3)}{8} P_{n-3}^{(3,3)}(t).
\end{aligned}$$

Thus, for $n \ge 3$,

$$P_n'''(t) = \frac{(n+1)(n+2)(n+3)}{8} P_{n-3}^{(3,3)}(t). \tag{7}$$

Taking $s = n-3$ and $\alpha = \beta = 3$ in the third estimate of that lemma yields

$$\left|P_{n-3}^{(3,3)}(t)\right| \le P_{n-3}^{(3,3)}(1) = \frac{(4)_{n-3}}{(n-3)!} = \frac{n(n-1)(n-2)}{6},$$

so (7) is at most $Cn^6$ uniformly on $[-1,1]$. Taking the same values in the fourth estimate, and absorbing the finitely many small values of $n$, gives

$$\left|P_{n-3}^{(3,3)}(\cos\theta)\right| \le Cn^{-1/2} \left(\sin\frac{\theta}{2}\right)^{-7/2} \left(\cos\frac{\theta}{2}\right)^{-7/2} \le Cn^{-1/2}(\sin\theta)^{-7/2}.$$

Combining this with (7) gives the second term in the minimum in (6). The cases $n < 3$ follow after enlarging $C$. □

## 2 Main result

**Theorem 2.1** (Sharp separated-support lower bound)**.** *There are absolute constants $C, c > 0$ such that the following holds. For every pair of integers*

$$1 \le \ell \le r$$

*and every real number*

$$d \ge C\ell r,$$

*there exist an $\ell$-Fourier-sparse signal $x$ and an $r$-Fourier-sparse signal $y$ such that*

$$\operatorname{dist}\big(\operatorname{supp}\widehat{x}, \operatorname{supp}\widehat{y}\big) \geq d$$

*and*

$$\Big|\langle x, y\rangle_{[-1,1]}\Big| \geq c\frac{\ell r}{d}\,\|x\|_{L^2[-1,1]}\,\|y\|_{L^2[-1,1]}\,.$$

The factor $\ell r/d$ is dimensionless because the observation interval has been normalized to $[-1,1]$. On $[-T,T]$, the corresponding factor is $\ell r/(dT)$.

The proof has two stages. Section 3 transfers a polynomial interaction bound to Fourier-sparse signals. Section 5 proves the sharp polynomial bound. Section 4 gives a shorter argument under the stronger assumption $d \gtrsim \ell^2 + r^2$.

## 3 From polynomials to Fourier sums

The following elementary lemma allows us to prove Theorem 2.1 entirely at the level of algebraic polynomials.

**Lemma 3.1** (Confluent realization)**.** *Let $\ell, r \geq 1$ be integers, let $D > d > 0$, and suppose that $p \in \mathcal{P}_\ell$ and $q \in \mathcal{P}_r$. For every $\eta > 0$, there are an $\ell$-Fourier-sparse signal $x$ and an $r$-Fourier-sparse signal $y$ such that*

$$\operatorname{dist}\big(\operatorname{supp}\widehat{x}, \operatorname{supp}\widehat{y}\big) \geq d,$$

$$|\|x\|_2 - \|p\|_2| \leq \eta, \qquad |\|y\|_2 - \|q\|_2| \leq \eta,$$

*and*

$$\left|\langle x, y\rangle_{[-1,1]} - \int_{-1}^{1} p(t)\overline{q(t)}\mathrm{e}^{-2\pi\mathrm{i}Dt}\,dt\right| \leq \eta.$$

*Proof.* For $\varepsilon > 0$, define

$$z_\varepsilon(t) := \frac{\mathrm{e}^{2\pi\mathrm{i}\varepsilon t} - 1}{2\pi\mathrm{i}\varepsilon}.$$

The standard limit $(\mathrm{e}^w - 1)/w \to 1$ as $w \to 0$ is uniform when $w = 2\pi\mathrm{i}\varepsilon t$ and $t \in [-1,1]$. Hence

$$\sup_{t\in[-1,1]} |z_\varepsilon(t) - t| \longrightarrow 0 \qquad (\varepsilon \to 0). \tag{8}$$

Set

$$x_\varepsilon(t) := p\big(z_\varepsilon(t)\big), \qquad y_\varepsilon(t) := \mathrm{e}^{2\pi\mathrm{i}Dt} q\big(z_\varepsilon(t)\big).$$

Write $p(t) = \sum_{h=0}^{\ell-1} c_h t^h$. The binomial theorem gives

$$z_\varepsilon(t)^h = \frac{1}{(2\pi\mathrm{i}\varepsilon)^h} \sum_{j=0}^{h} \binom{h}{j} (-1)^{h-j} \mathrm{e}^{2\pi\mathrm{i}j\varepsilon t}.$$

Therefore

$$x_\varepsilon \in \operatorname{span}\{\mathrm{e}^{2\pi\mathrm{i}j\varepsilon t} : 0 \leq j < \ell\}.$$

The same calculation for $q$ gives

$$y_\varepsilon \in \operatorname{span}\{\mathrm{e}^{2\pi\mathrm{i}(D+k\varepsilon)t} : 0 \leq k < r\}.$$

Thus $x_\varepsilon$ and $y_\varepsilon$ are respectively $\ell$- and $r$-Fourier-sparse according to the convention of Section 1. Every frequency of $x_\varepsilon$ belongs to $[0, (\ell-1)\varepsilon]$, whereas every frequency of $y_\varepsilon$ belongs to $[D, D+(r-1)\varepsilon]$. Their frequency supports are therefore separated by at least

$$D - (\ell - 1)\varepsilon.$$

Since $D > d$, this quantity is at least $d$ whenever

$$0 < \varepsilon \leq \frac{D-d}{\max\{1, \ell-1\}}.$$

By (8) and uniform continuity of the fixed polynomials on a compact neighborhood of $[-1, 1]$,

$$x_\varepsilon \to p, \qquad y_\varepsilon \to \mathrm{e}^{2\pi \mathrm{i} D t} q$$

uniformly on $[-1, 1]$. In particular, both convergences hold in $L^2$, and the reverse triangle inequality gives

$$\begin{aligned} |\|x_\varepsilon\|_2 - \|p\|_2| &\leq \|x_\varepsilon - p\|_2 \longrightarrow 0, \\ |\|y_\varepsilon\|_2 - \|q\|_2| &= \left|\|y_\varepsilon\|_2 - \left\|\mathrm{e}^{2\pi \mathrm{i} D t} q\right\|_2\right| \\ &\leq \left\|y_\varepsilon - \mathrm{e}^{2\pi \mathrm{i} D t} q\right\|_2 \longrightarrow 0. \end{aligned}$$

For the inner product, Cauchy–Schwarz gives

$$\begin{aligned} &\left|\langle x_\varepsilon, y_\varepsilon\rangle_{[-1,1]} - \left\langle p, \mathrm{e}^{2\pi \mathrm{i} D t} q\right\rangle_{[-1,1]}\right| \\ &\quad \leq \|x_\varepsilon - p\|_2 \, \|y_\varepsilon\|_2 + \|p\|_2 \left\|y_\varepsilon - \mathrm{e}^{2\pi \mathrm{i} D t} q\right\|_2 \longrightarrow 0. \end{aligned}$$

Finally,

$$\left\langle p, \mathrm{e}^{2\pi \mathrm{i} D t} q\right\rangle_{[-1,1]} = \int_{-1}^{1} p(t)\overline{q(t)}\mathrm{e}^{-2\pi \mathrm{i} D t}\, dt.$$

Choose $\varepsilon > 0$ small enough to satisfy the separation condition and all three approximation bounds. □

*Remark* 3.2. The construction gives sparsity "at most" $\ell$ and $r$. If exactly $\ell$ and $r$ nonzero atoms are desired, one may add arbitrarily small atoms at unused frequencies inside the two clusters; all inequalities persist after reducing the absolute constant.

## 4 A simpler endpoint-kernel proof

This section proves the desired lower bound under the stronger separation assumption

$$d \gtrsim \ell^2 + r^2.$$

In particular, it already proves Theorem 2.1 whenever $\ell$ and $r$ are comparable.

Define the endpoint Christoffel–Darboux kernel

$$K_m(t) := \sum_{j=0}^{m-1} \frac{2j+1}{2} P_j(t), \qquad u_m(t) := \frac{\sqrt{2}}{m} K_m(t).$$

**Lemma 4.1** (Endpoint kernels)**.** *For every integer $m \geq 1$,*

$$\|u_m\|_2 = 1, \qquad u_m(1) = \frac{m}{\sqrt{2}}, \qquad u_m(-1) = \frac{(-1)^{m-1}}{\sqrt{2}}.$$

*Moreover,*

$$\|u_m\|_\infty \lesssim m, \qquad \|u_m'\|_2 \lesssim m^2, \qquad \|u_m''\|_1 \lesssim m^3.$$

*Proof.* By (1), all cross terms in the squared norm of $K_m$ vanish. Therefore

$$\begin{aligned}\|K_m\|_2^2 &= \sum_{j=0}^{m-1} \left(\frac{2j+1}{2}\right)^2 \frac{2}{2j+1} \\ &= \frac{1}{2}\sum_{j=0}^{m-1}(2j+1) = \frac{m^2}{2}.\end{aligned}$$

Since $u_m = (\sqrt{2}/m)K_m$, this proves $\|u_m\|_2 = 1$.

Next, (2) gives

$$K_m(1) = \frac{m^2}{2}.$$

At the other endpoint, the finite sum

$$\sum_{j=0}^{m-1}(2j+1)(-1)^j = (-1)^{m-1}m,$$

gives

$$K_m(-1) = \frac{(-1)^{m-1}m}{2}.$$

Multiplication by $\sqrt{2}/m$ proves both endpoint identities.

For the supremum bound, Cauchy–Schwarz and (3) give, for each $t \in [-1,1]$,

$$\begin{aligned}|K_m(t)|^2 &\leq \left(\sum_{j=0}^{m-1}\frac{2j+1}{2}\right)\left(\sum_{j=0}^{m-1}\frac{2j+1}{2}P_j(t)^2\right) \\ &\leq \frac{m^2}{2}\cdot\frac{m^2}{2}.\end{aligned}$$

Consequently, $\|K_m\|_\infty \leq m^2/2$ and $\|u_m\|_\infty \leq m/\sqrt{2}$.

Applying (5) to $u_m$, whose degree is at most $m-1$, gives

$$\|u_m'\|_2 \lesssim (m-1)^2\,\|u_m\|_2 \leq m^2.$$

Finally, (4) implies

$$K_m(t) = \frac{1}{2}\big(P_m'(t) + P_{m-1}'(t)\big),$$

so

$$u_m''(t) = \frac{P_m'''(t) + P_{m-1}'''(t)}{\sqrt{2}\,m}.$$

For $m = 1$, the function $u_m$ is constant and the asserted bound is immediate. Suppose henceforth that $m \geq 2$. To derive the required $L^1$ bound from (6), symmetry about $\pi/2$ and the change of variables $t = \cos\theta$ give

$$\left\|P_n'''\right\|_{L^1[-1,1]} \lesssim \int_0^{1/n} n^6 \sin\theta \, d\theta + \int_{1/n}^{\pi/2} n^{5/2} (\sin\theta)^{-5/2} \, d\theta.$$

Since $\sin\theta \asymp \theta$ on $[0, \pi/2]$,

$$\int_0^{1/n} n^6 \sin\theta \, d\theta \lesssim n^6 \int_0^{1/n} \theta \, d\theta \lesssim n^4,$$
$$\int_{1/n}^{\pi/2} n^{5/2} (\sin\theta)^{-5/2} \, d\theta \lesssim n^{5/2} \int_{1/n}^{\pi/2} \theta^{-5/2} \, d\theta \lesssim n^4.$$

Therefore $\|P_n'''\|_{L^1[-1,1]} \lesssim n^4$. Substitution for $n = m$ and $n = m - 1$ in the formula for $u_m''$ yields $\|u_m''\|_1 \lesssim m^3$. □

**Proposition 4.2** (Elementary large-separation lower bound)**.** *There are absolute constants $C, c > 0$ such that the following holds. Let $1 \leq \ell \leq r$ with $\ell r \geq 2$, let $N$ be a positive integer, and suppose*

$$N \geq C(\ell^2 + r^2).$$

*Then there are unit vectors $p \in \mathcal{P}_\ell$ and $q \in \mathcal{P}_r$ such that*

$$\left| \int_{-1}^{1} p(t)\overline{q(t)} \mathrm{e}^{-2\pi \mathrm{i} N t} \, dt \right| \geq c \frac{\ell r}{N}.$$

*Proof.* Take

$$p = u_\ell, \qquad q = u_r.$$

These polynomials are real and have unit $L^2$ norm. Put $h = pq$ and $\omega = 2\pi N$. By Lemma 4.1,

$$h(1) = \frac{\ell r}{2}, \qquad h(-1) = \frac{(-1)^{\ell + r}}{2},$$

and hence, because $\ell r \geq 2$,

$$h(1) - h(-1) \geq \frac{\ell r - 1}{2} \geq \frac{\ell r}{4}.$$

We next bound the error term that arises after two integrations by parts. The product rule gives

$$h'' = p''q + 2p'q' + pq'',$$

so Hölder's inequality, followed by Lemma 4.1, gives

$$\begin{aligned} \left\|h''\right\|_1 &\leq \left\|p''\right\|_1 \|q\|_\infty + 2 \left\|p'\right\|_2 \left\|q'\right\|_2 + \|p\|_\infty \left\|q''\right\|_1 \\ &\lesssim \ell^3 r + \ell^2 r^2 + \ell r^3 \\ &= \ell r(\ell^2 + \ell r + r^2) \\ &\leq 2\ell r(\ell^2 + r^2). \end{aligned}$$

Let

$$I := \int_{-1}^{1} h(t) \mathrm{e}^{-\mathrm{i}\omega t} \, dt.$$

Because $N$ is an integer, $\mathrm{e}^{\pm \mathrm{i}\omega} = 1$. Integrating twice by parts gives

$$\begin{aligned} I &= \frac{\mathrm{i}}{\omega}\big[h(t)\mathrm{e}^{-\mathrm{i}\omega t}\big]_{-1}^{1} - \frac{\mathrm{i}}{\omega}\int_{-1}^{1} h'(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt \\ &= \frac{\mathrm{i}\big(h(1)-h(-1)\big)}{\omega} + \frac{1}{\omega^2}\big[h'(t)\mathrm{e}^{-\mathrm{i}\omega t}\big]_{-1}^{1} - \frac{1}{\omega^2}\int_{-1}^{1} h''(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt \\ &= \frac{\mathrm{i}\big(h(1)-h(-1)\big)}{\omega} + \frac{h'(1)-h'(-1)}{\omega^2} - \frac{1}{\omega^2}\int_{-1}^{1} h''(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt. \end{aligned}$$

Since $h$ is real-valued, $h'(1) - h'(-1)$ is real. Taking imaginary parts and using $|I| \geq |\operatorname{Im} I|$ therefore yields

$$|I| \geq \frac{h(1)-h(-1)}{\omega} - \frac{\|h''\|_1}{\omega^2}.$$

The two preceding estimates and $\omega = 2\pi N$ show

$$\begin{aligned} |I| &\geq \frac{\ell r}{4\omega} - C\frac{\ell r(\ell^2+r^2)}{\omega^2} \\ &= \frac{\ell r}{\omega}\left(\frac{1}{4} - C\frac{\ell^2+r^2}{\omega}\right). \end{aligned}$$

If $N \geq C_0(\ell^2 + r^2)$ for a sufficiently large absolute constant $C_0$, the parenthesis is at least $1/8$. Hence $|I| \gtrsim \ell r/N$.

□

**Corollary 4.3** (Signal lower bound in the simpler regime)**.** *Let $1 \leq \ell \leq r$ be integers. If*

$$d \geq C(\ell^2 + r^2),$$

*then there are separated $\ell$- and $r$-Fourier-sparse signals satisfying*

$$\left|\langle x, y\rangle_{[-1,1]}\right| \gtrsim \frac{\ell r}{d}\,\|x\|_2\,\|y\|_2\,.$$

*Proof.* If $\ell r \geq 2$, choose an integer $N \in [d+1, d+2]$. Proposition 4.2 gives unit polynomials $p$ and $q$ such that

$$\left|\int_{-1}^{1} p(t)\overline{q(t)}\mathrm{e}^{-2\pi\mathrm{i}Nt}\,dt\right| \geq c_0\frac{\ell r}{N}.$$

Apply Lemma 3.1 with $D = N$ and choose its approximation parameter so that

$$\eta \leq \min\left\{\frac{1}{2}, \frac{c_0\ell r}{4N}\right\}.$$

Then

$$\begin{aligned} \left|\langle x, y\rangle_{[-1,1]}\right| &\geq \frac{3c_0\ell r}{4N}, \\ \|x\|_2\,\|y\|_2 &\leq (1+\eta)^2 \leq \frac{9}{4}. \end{aligned}$$

Since $N \leq d + 2 \lesssim d$ under the stated hypothesis, these inequalities give the claimed bound.

If $\ell = r = 1$, choose $D \in [d+1, d+2]$ with fractional part $1/4$ and take $p = q = 1$. Direct integration gives

$$\left|\int_{-1}^{1} \mathrm{e}^{-2\pi \mathrm{i} D t}\, dt\right| = \left|\frac{\mathrm{e}^{-2\pi \mathrm{i} D} - \mathrm{e}^{2\pi \mathrm{i} D}}{-2\pi \mathrm{i} D}\right| = \frac{|\sin(2\pi D)|}{\pi D} = \frac{1}{\pi D} \gtrsim \frac{1}{d}.$$

Choose $\eta$ in Lemma 3.1 small enough that the resulting inner product is at least half this value and both norms are at most $2\,\|1\|_2$. The asserted normalized lower bound follows. □

# 5 The sharp polynomial interaction bound

We now remove the stronger condition $d \gtrsim \ell^2 + r^2$. The key point is to retain the order-dependent phase of the Fourier transform of a Legendre polynomial.

Let

$$\phi_n(t) := \sqrt{\frac{2n+1}{2}} P_n(t),$$

so that $\{\phi_n\}_{n \geq 0}$ is orthonormal in $L^2[-1, 1]$.

## 5.1 A uniform oscillatory formula

We first state every imported Bessel and Legendre-transform formula used in the oscillatory estimate. The subsequent lemmas then perform the required substitutions explicitly.

**Lemma 5.1** (Finite Legendre transform)**.** *For every integer $n \geq 0$ and every $\omega > 0$, the formulas used here are the following, where $J_\nu$ is the Bessel function of the first kind and $j_n$ is the spherical Bessel function of the first kind:*

$$\int_{-1}^{1} P_n(t) \mathrm{e}^{\mathrm{i}\omega t}\, dt = \mathrm{i}^n \sqrt{\frac{2\pi}{\omega}} J_{n+1/2}(\omega), \tag{DLMF 18.17.19}$$

$$j_n(\omega) = \sqrt{\frac{\pi}{2\omega}} J_{n+1/2}(\omega). \tag{DLMF 10.47.3}$$

*See [NIS, Eqs. (10.47.3) and (18.17.19)]. Consequently,*

$$\int_{-1}^{1} P_n(t) \mathrm{e}^{-\mathrm{i}\omega t}\, dt = 2(-\mathrm{i})^n j_n(\omega) = 2(-\mathrm{i})^n \sqrt{\frac{\pi}{2\omega}} J_{n+1/2}(\omega). \tag{9}$$

*Proof.* Since $P_n$ and $J_{n+1/2}$ are real on the indicated real arguments, complex conjugation of the first displayed formula gives

$$\int_{-1}^{1} P_n(t) \mathrm{e}^{-\mathrm{i}\omega t}\, dt = (-\mathrm{i})^n \sqrt{\frac{2\pi}{\omega}} J_{n+1/2}(\omega). \tag{10}$$

The second formula gives

$$\sqrt{\frac{2\pi}{\omega}} J_{n+1/2}(\omega) = 2\sqrt{\frac{\pi}{2\omega}} J_{n+1/2}(\omega) = 2 j_n(\omega).$$

Substitution into (10) proves (9). □

**Lemma 5.2** (Bessel asymptotics)**.** *Let $J_\nu$ denote the Bessel function of the first kind of order $\nu$. For $\nu > 0$ and $0 < \beta < \pi/2$, set*

$$\xi := \nu(\tan\beta - \beta) - \frac{\pi}{4}.$$

*For every fixed $\delta > 0$, the first term of the oscillatory Debye expansion, together with its remainder, has the form*

$$J_\nu(\nu \sec\beta) = \left(\frac{2}{\pi\nu\tan\beta}\right)^{1/2} \big(\cos\xi + E_\nu(\beta)\big), \qquad |E_\nu(\beta)| \le \frac{C_\delta}{\nu\sec\beta}. \tag{11}$$

*This holds uniformly for $\sec\beta \ge 1 + \delta$ once $\nu$ exceeds a constant depending only on $\delta$; $C_\delta$ depends only on $\delta$. The phase and principal term are [NIS, Eqs. (10.19.5)–(10.19.6)], and the displayed remainder is the corresponding error bound in [Olv97, p. 382].*

*For every fixed order $\nu$, the large-argument formula with its real-argument remainder is*

$$J_\nu(\omega) = \sqrt{\frac{2}{\pi\omega}}\left(\cos\left(\omega - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + r_\nu(\omega)\right), \qquad |r_\nu(\omega)| \le \frac{C_\nu}{\omega} \tag{12}$$

*for all sufficiently large $\omega$, where $C_\nu$ depends only on $\nu$. See [NIS, Eq. (10.17.3) and Sec. 10.17(iii)] and [Olv97, pp. 268–269].*

**Lemma 5.3** (Uniform Debye–Olver expansion)**.** *There are absolute constants $\omega_0, C > 0$ such that, if $\omega \ge \omega_0$ and $\nu_n := n + 1/2 \le \omega/4$, then*

$$j_n(\omega) = \frac{1}{\omega}\left(1 - \frac{\nu_n^2}{\omega^2}\right)^{-1/4} \left(\cos\Xi_n(\omega) + \varepsilon_n(\omega)\right), \tag{13}$$

*where*

$$\Xi_n(\omega) = \sqrt{\omega^2 - \nu_n^2} - \nu_n \arccos\frac{\nu_n}{\omega} - \frac{\pi}{4}$$

*and $|\varepsilon_n(\omega)| \le C/\omega$ uniformly in $n$ and $\omega$.*

*Proof.* We first apply (11) from Lemma 5.2. Put $\nu = \nu_n$ and choose $\beta$ by

$$\sec\beta = \frac{\omega}{\nu_n}.$$

The assumption $\nu_n \le \omega/4$ gives $\sec\beta \ge 4$, so (11) applies with an absolute constant. Moreover,

$$\cos\beta = \frac{\nu_n}{\omega}, \qquad \nu_n\tan\beta = \sqrt{\omega^2 - \nu_n^2}, \qquad \beta = \arccos\frac{\nu_n}{\omega}.$$

Consequently, the cited phase becomes

$$\begin{aligned}\xi &= \nu_n\tan\beta - \nu_n\beta - \frac{\pi}{4}\\ &= \sqrt{\omega^2 - \nu_n^2} - \nu_n\arccos\frac{\nu_n}{\omega} - \frac{\pi}{4}\\ &= \Xi_n(\omega).\end{aligned}$$

Using the spherical-Bessel identity displayed in Lemma 5.1, the prefactor becomes

$$\begin{aligned}\sqrt{\frac{\pi}{2\omega}}\left(\frac{2}{\pi\nu_n\tan\beta}\right)^{1/2} &= \frac{1}{\sqrt{\omega\sqrt{\omega^2 - \nu_n^2}}}\\ &= \frac{1}{\omega}\left(1 - \frac{\nu_n^2}{\omega^2}\right)^{-1/4}.\end{aligned}$$

Finally,

$$|E_{\nu_n}(\beta)| \le \frac{C}{\nu_n \sec\beta} = \frac{C}{\omega}.$$

Taking $\delta = 3$ proves (13) for all sufficiently large orders.

For each of the finitely many remaining bounded orders, use (12). Elementary Taylor estimates give a constant $C_n$ such that

$$\left|\Xi_n(\omega) - \left(\omega - \frac{\nu_n\pi}{2} - \frac{\pi}{4}\right)\right| \le \frac{C_n}{\omega}, \qquad \left|\left(1 - \frac{\nu_n^2}{\omega^2}\right)^{-1/4} - 1\right| \le \frac{C_n}{\omega^2}.$$

Write

$$a_n := \left(1 - \frac{\nu_n^2}{\omega^2}\right)^{-1/4}.$$

Multiplying (12) by $\sqrt{\pi/(2\omega)}$ gives

$$j_n(\omega) = \frac{1}{\omega}\left(\cos\left(\omega - \frac{\nu_n\pi}{2} - \frac{\pi}{4}\right) + r_{\nu_n}(\omega)\right) = \frac{a_n}{\omega}\big(\cos\Xi_n(\omega) + \varepsilon_n(\omega)\big),$$

where

$$\varepsilon_n := a_n^{-1}\left(\cos\left(\omega - \frac{\nu_n\pi}{2} - \frac{\pi}{4}\right) + r_{\nu_n}(\omega)\right) - \cos\Xi_n.$$

Since $|\cos u - \cos v| \le |u - v|$ and $a_n$ is bounded above and below, the two Taylor estimates and (12) give

$$|\varepsilon_n| \le \left|a_n^{-1} - 1\right| + \left|\omega - \frac{\nu_n\pi}{2} - \frac{\pi}{4} - \Xi_n\right| + a_n^{-1}\,|r_{\nu_n}(\omega)| \le \frac{C_n}{\omega}.$$

Increasing $C$ and $\omega_0$ over this finite set of orders proves (13) in all remaining cases. □

The next lemma extracts the exact phase and amplitude estimates used later.

**Lemma 5.4** (Uniform Legendre-transform asymptotic)**.** *There are absolute constants $\omega_0, C > 0$ such that the following holds. For $\omega \ge \omega_0$ and every integer $n \ge 0$ satisfying*

$$\nu_n := n + \frac{1}{2} \le \frac{\omega}{4},$$

*set*

$$\Xi_n(\omega) := \sqrt{\omega^2 - \nu_n^2} - \nu_n \arccos\frac{\nu_n}{\omega} - \frac{\pi}{4},$$

$$a_n(\omega) := \left(1 - \frac{\nu_n^2}{\omega^2}\right)^{-1/4},$$

*and*

$$\theta_n(\omega) := \omega - \Xi_n(\omega) - \frac{n\pi}{2}.$$

*Then*

$$F_n(\omega) := \int_{-1}^{1} P_n(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt = \frac{a_n}{\omega}\left((-1)^n\mathrm{e}^{\mathrm{i}\omega - \mathrm{i}\theta_n} + \mathrm{e}^{-\mathrm{i}\omega + \mathrm{i}\theta_n}\right) + \rho_n, \tag{14}$$

*where*

$$|\rho_n| \le \frac{C}{\omega^2}.$$

*Furthermore, whenever $m, n$ satisfy the same size condition,*

$$|\theta_m - \theta_n| \le C\frac{|m-n|\,(m+n+1)}{\omega}, \tag{15}$$

*and*

$$|a_m - a_n| \le C\frac{|m-n|\,(m+n+1)}{\omega^2}. \tag{16}$$

*Also* $1 \le a_n \le 2$.

*Proof.* By Lemmas 5.1 and 5.3,

$$\begin{aligned} F_n(\omega) &= \frac{2a_n}{\omega}(-\mathrm{i})^n\left(\cos\Xi_n + \varepsilon_n\right) \\ &= \frac{a_n}{\omega}(-\mathrm{i})^n\left(\mathrm{e}^{\mathrm{i}\Xi_n} + \mathrm{e}^{-\mathrm{i}\Xi_n}\right) + \rho_n, \end{aligned}$$

where

$$|\rho_n| \le \frac{2a_n}{\omega}\,|\varepsilon_n| \lesssim \frac{1}{\omega^2}.$$

The definition of $\theta_n$ is equivalent to $\Xi_n = \omega - \theta_n - n\pi/2$. Consequently,

$$\begin{aligned} (-\mathrm{i})^n\mathrm{e}^{\mathrm{i}\Xi_n} &= \mathrm{e}^{-\mathrm{i}n\pi/2}\mathrm{e}^{\mathrm{i}\omega-\mathrm{i}\theta_n-\mathrm{i}n\pi/2} = (-1)^n\mathrm{e}^{\mathrm{i}\omega-\mathrm{i}\theta_n}, \\ (-\mathrm{i})^n\mathrm{e}^{-\mathrm{i}\Xi_n} &= \mathrm{e}^{-\mathrm{i}n\pi/2}\mathrm{e}^{-\mathrm{i}\omega+\mathrm{i}\theta_n+\mathrm{i}n\pi/2} = \mathrm{e}^{-\mathrm{i}\omega+\mathrm{i}\theta_n}. \end{aligned}$$

Substitution proves (14).

It remains to prove the variation estimates. Regard $\nu$ as a continuous variable. Direct differentiation gives

$$\begin{aligned} \frac{\partial}{\partial\nu}\left(\sqrt{\omega^2-\nu^2} - \nu\arccos\frac{\nu}{\omega}\right) &= -\frac{\nu}{\sqrt{\omega^2-\nu^2}} - \arccos\frac{\nu}{\omega} + \frac{\nu}{\sqrt{\omega^2-\nu^2}} \\ &= -\arccos\frac{\nu}{\omega}. \end{aligned}$$

Since $n = \nu - 1/2$,

$$\frac{\partial\theta}{\partial\nu} = \arccos\frac{\nu}{\omega} - \frac{\pi}{2} = -\arcsin\frac{\nu}{\omega}.$$

For $0 \le \nu \le \omega/4$, the elementary bound $\arcsin(\nu/\omega) \le C\nu/\omega$ and the mean value theorem give

$$\begin{aligned} |\theta_m - \theta_n| &\le \frac{C}{\omega}\,|\nu_m - \nu_n|\max\{\nu_m, \nu_n\} \\ &\le C\frac{|m-n|\,(m+n+1)}{\omega}, \end{aligned}$$

which is (15). Likewise,

$$\left|\frac{\partial a}{\partial\nu}\right| = \frac{\nu}{2\omega^2}\left(1 - \frac{\nu^2}{\omega^2}\right)^{-5/4} \le C\frac{\nu}{\omega^2},$$

and the mean value theorem gives

$$|a_m - a_n| \le C\frac{|m-n|\,(m+n+1)}{\omega^2},$$

which is (16). Finally, $0 \le \nu_n/\omega \le 1/4$ implies

$$1 \le a_n \le (1 - 1/16)^{-1/4} < 2.$$

□

### 5.2 Positive product linearization

The following imported identity supplies the positivity used in the coherent summation argument.

**Lemma 5.5** (Legendre product formula)**.** *For nonnegative integers $j, k$ and each integer $n$ satisfying $|j-k| \le n \le j+k$ and $j+k+n$ even, define the integer*

$$s_n := \frac{j+k+n}{2}.$$

*DLMF Eq. (34.3.19) is a Legendre product formula. Writing out its coefficient by means of DLMF Eq. (34.3.5) gives the following identity using only Legendre polynomials, square roots, and factorials:*

$$P_j(t)P_k(t) = \sum_{\substack{|j-k|\le n\le j+k \\ j+k+n \text{ even}}} (2n+1) \cdot \left[(-1)^{s_n} \frac{s_n!}{(s_n-j)!(s_n-k)!(s_n-n)!} \left(\frac{(2s_n-2j)!(2s_n-2k)!(2s_n-2n)!}{(2s_n+1)!}\right)^{1/2}\right]^2 P_n(t). \tag{17}$$

*See [NIS, Eqs. (34.3.5) and (34.3.19)]. Equation* (17) *writes out the source's specialized coupling coefficient and therefore contains no undeclared coupling symbol.*

**Lemma 5.6** (Positive Legendre product linearization)**.** *For integers $j, k \ge 0$,*

$$P_j(t)P_k(t) = \sum_{n=|j-k|}^{j+k} \gamma_{j,k,n} P_n(t), \tag{18}$$

*Here $\gamma_{j,k,n}$ is the scalar coefficient multiplying $P_n(t)$ in this Legendre-basis expansion of the product $P_j(t)P_k(t)$. If $|j-k| \le n \le j+k$ and $j+k+n$ is even, then the coefficient is given entirely in terms of factorials by*

$$\begin{aligned}\gamma_{j,k,n} &= \frac{2n+1}{2}\int_{-1}^{1} P_j(t)P_k(t)P_n(t)\,dt \\ &= (2n+1)\frac{(j+k-n)!(j-k+n)!(-j+k+n)!}{(j+k+n+1)!} \\ &\quad \cdot \left(\frac{\big((j+k+n)/2\big)!}{\big((j+k-n)/2\big)!\big((j-k+n)/2\big)!\big((-j+k+n)/2\big)!}\right)^2.\end{aligned} \tag{19}$$

*All factorial arguments in* (19) *are then nonnegative integers. If the triangle condition fails or $j+k+n$ is odd, set $\gamma_{j,k,n} := 0$. The factorial formula makes positivity explicit. In particular,*

$$\gamma_{j,k,n} \ge 0, \qquad \gamma_{j,k,n} = 0 \text{ unless } n \equiv j+k \pmod 2, \qquad \sum_n \gamma_{j,k,n} = 1.$$

*Proof.* The cited product identity (17) first gives the expansion (18); it also says immediately that the coefficient is zero unless the triangle and parity conditions hold.

Multiplying by $P_n$ and integrating, the orthogonality relation (1) gives

$$\int_{-1}^{1} P_j(t)P_k(t)P_n(t)\,dt = \frac{2}{2n+1}\gamma_{j,k,n},$$

which proves the first equality in (19).

To rewrite this identity without skipping the change of variables, write $s = s_n = (j+k+n)/2$. Then

$$\begin{aligned} s-j &= \frac{-j+k+n}{2}, & s-k &= \frac{j-k+n}{2}, & s-n &= \frac{j+k-n}{2}, \\ 2s-2j &= -j+k+n, & 2s-2k &= j-k+n, & 2s-2n &= j+k-n, \\ 2s+1 &= j+k+n+1. \end{aligned}$$

The square in (17) removes the factor $(-1)^s$. Substituting the seven displayed equalities into the coefficient of $P_n$ in that identity gives

$$\begin{aligned} \gamma_{j,k,n} = (2n+1)&\frac{(j+k-n)!(j-k+n)!(-j+k+n)!}{(j+k+n+1)!} \\ &\cdot \left( \frac{\big((j+k+n)/2\big)!}{\big((j+k-n)/2\big)!\big((j-k+n)/2\big)!\big((-j+k+n)/2\big)!} \right)^2, \end{aligned}$$

which is the second equality in (19). The zero cases follow from the zero cases in Lemma 5.5.

It remains only to verify the normalization used below. Setting $t = 1$ in (18) and using $P_m(1) = 1$ for every $m \geq 0$ gives

$$1 = P_j(1)P_k(1) = \sum_n \gamma_{j,k,n} P_n(1) = \sum_n \gamma_{j,k,n}.$$

□

**Lemma 5.7** (Coherent transform of a Legendre product)**.** *There are absolute constants $C_0, C > 0$ with the following property. Suppose*

$$1 \leq \ell \leq r, \qquad r \geq 2, \qquad \omega \geq C_0 \ell r,$$

*where $\ell$ and $r$ are integers. If*

$$0 \leq j < \ell_0, \qquad k \in \mathcal{K},$$

*where*

$$\ell_0 := \min\left\{\ell, \left\lfloor \frac{r}{2} \right\rfloor\right\}, \qquad \mathcal{K} := \left\{\left\lceil \frac{r}{2} \right\rceil, \ldots, r-1\right\}.$$

*Then $j \leq k$, and*

$$\int_{-1}^{1} P_j(t)P_k(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt = \frac{a_k}{\omega}\left((-1)^{j+k}\mathrm{e}^{\mathrm{i}\omega - \mathrm{i}\theta_k} + \mathrm{e}^{-\mathrm{i}\omega + \mathrm{i}\theta_k}\right) + R_{jk}, \tag{20}$$

*where*

$$|R_{jk}| \leq C\frac{(j+1)(k+1)}{\omega^2}. \tag{21}$$

*Proof.* The definitions of $\ell_0$ and $\mathcal{K}$ imply

$$j \leq \left\lfloor \frac{r}{2} \right\rfloor - 1 < \left\lceil \frac{r}{2} \right\rceil \leq k.$$

By Lemma 5.6, every index with $\gamma_{j,k,n} \neq 0$ therefore satisfies

$$k - j \leq n \leq k + j, \qquad (-1)^n = (-1)^{j+k}.$$

Also

$$n+\frac{1}{2}\le j+k+\frac{1}{2}<\ell+r\le 2\ell r.$$

Thus $n+1/2\le\omega/4$ after taking $C_0\ge 8$, and Lemma 5.4 applies. The same index range gives

$$|n-k|\le j,\qquad n+k+1\le j+2k+1\le 3(k+1),$$

where the last inequality uses $j\le k$. Hence (15) and (16) imply

$$|\theta_n-\theta_k|\lesssim\frac{j(k+1)}{\omega}\le\frac{(j+1)(k+1)}{\omega},$$

and

$$|a_n-a_k|\lesssim\frac{j(k+1)}{\omega^2}\le\frac{(j+1)(k+1)}{\omega^2}.$$

We now calculate the transform of the product. By Lemma 5.6 and (14),

$$\begin{aligned}\int_{-1}^{1}P_j(t)P_k(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt&=\sum_n\gamma_{j,k,n}F_n(\omega)\\&=\frac{1}{\omega}\sum_n\gamma_{j,k,n}a_n\left((-1)^{j+k}\mathrm{e}^{\mathrm{i}\omega-\mathrm{i}\theta_n}+\mathrm{e}^{-\mathrm{i}\omega+\mathrm{i}\theta_n}\right)+\sum_n\gamma_{j,k,n}\rho_n.\end{aligned}$$

For either choice of sign, the triangle inequality and $\left|\mathrm{e}^{\mathrm{i}s}-\mathrm{e}^{\mathrm{i}t}\right|\le|s-t|$ give

$$\begin{aligned}\left|a_n\mathrm{e}^{\pm\mathrm{i}\theta_n}-a_k\mathrm{e}^{\pm\mathrm{i}\theta_k}\right|&\le|a_n-a_k|+a_k\left|\mathrm{e}^{\pm\mathrm{i}\theta_n}-\mathrm{e}^{\pm\mathrm{i}\theta_k}\right|\\&\le|a_n-a_k|+2\,|\theta_n-\theta_k|\\&\lesssim\frac{(j+1)(k+1)}{\omega}.\end{aligned}$$

Because $\gamma_{j,k,n}\ge 0$ and $\sum_n\gamma_{j,k,n}=1$, replacing every $a_n\mathrm{e}^{\pm\mathrm{i}\theta_n}$ by $a_k\mathrm{e}^{\pm\mathrm{i}\theta_k}$ therefore contributes at most

$$\frac{C(j+1)(k+1)}{\omega^2}.$$

The remainder contributes at most

$$\left|\sum_n\gamma_{j,k,n}\rho_n\right|\le\sum_n\gamma_{j,k,n}\,|\rho_n|\le\frac{C}{\omega^2}\le\frac{C(j+1)(k+1)}{\omega^2}.$$

Combining these bounds proves (20) and (21). □

### 5.3 The cross-Gram matrix

**Proposition 5.8** (Sharp polynomial interaction)**.** *There are absolute constants $C,c>0$ such that, whenever*

$$1\le\ell\le r,\qquad r\ge 2,\qquad\omega\ge C\ell r,$$

*there exist $p\in\mathcal{P}_\ell$ and $q\in\mathcal{P}_r$ with*

$$\|p\|_2=\|q\|_2=1$$

*and*

$$\left|\int_{-1}^{1}p(t)\overline{q(t)}\mathrm{e}^{-\mathrm{i}\omega t}\,dt\right|\ge c\frac{\ell r}{\omega}.$$

*Proof.* Let $\ell_0$ and $\mathcal{K}$ be as in Lemma 5.7. Since $r \geq 2$,

$$\left\lfloor \frac{r}{2} \right\rfloor \geq \frac{r}{3} \geq \frac{\ell}{3}.$$

Consequently,

$$\ell_0 \geq \frac{\ell}{3}. \tag{22}$$

Consider the $\ell_0 \times |\mathcal{K}|$ matrix

$$A_{jk} := \int_{-1}^{1} \phi_j(t)\phi_k(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt, \qquad 0 \leq j < \ell_0, \quad k \in \mathcal{K}.$$

This is a submatrix of the matrix of the operator

$$\Pi_\ell M_\omega \Pi_r, \qquad (M_\omega f)(t) := \mathrm{e}^{-\mathrm{i}\omega t} f(t),$$

where $\Pi_m$ is the orthogonal projection onto $\mathcal{P}_m$. Hence

$$\|\Pi_\ell M_\omega \Pi_r\|_{2\to 2} \geq \|A\|_{\mathrm{op}}.$$

By Lemma 5.7,

$$A = L + E,$$

where

$$L_{jk} = \frac{\sqrt{(2j+1)(2k+1)}}{2\omega} a_k \left( (-1)^{j+k} \mathrm{e}^{\mathrm{i}\omega - \mathrm{i}\theta_k} + \mathrm{e}^{-\mathrm{i}\omega + \mathrm{i}\theta_k} \right)$$

and

$$|E_{jk}| \leq \sqrt{\frac{(2j+1)(2k+1)}{4}}\,|R_{jk}| \lesssim \frac{(j+1)^{3/2}(k+1)^{3/2}}{\omega^2}.$$

Since $\|E\|_{\mathrm{op}} \leq \|E\|_{\mathrm{HS}}$ and

$$\sum_{j<\ell_0} (j+1)^3 \lesssim \ell_0^4, \qquad \sum_{k\in\mathcal{K}} (k+1)^3 \lesssim r^4,$$

we obtain

$$\begin{aligned} \|E\|_{\mathrm{op}} &\leq \|E\|_{\mathrm{HS}} \\ &\lesssim \frac{1}{\omega^2} \left( \sum_{j<\ell_0} (j+1)^3 \right)^{1/2} \left( \sum_{k\in\mathcal{K}} (k+1)^3 \right)^{1/2} \\ &\lesssim \frac{\ell_0^2 r^2}{\omega^2}. \end{aligned} \tag{23}$$

Define vectors indexed respectively by $0 \leq j < \ell_0$ and $k \in \mathcal{K}$:

$$u_j := \sqrt{\frac{2j+1}{2}}, \qquad u_j^- := (-1)^j u_j,$$

$$v_k := \sqrt{\frac{2k+1}{2}} a_k \mathrm{e}^{\mathrm{i}\theta_k}, \qquad v_k^- := (-1)^k \sqrt{\frac{2k+1}{2}} a_k \mathrm{e}^{-\mathrm{i}\theta_k}.$$

Then

$$L = \frac{1}{\omega}\left(\mathrm{e}^{-\mathrm{i}\omega} u v^{\mathsf{T}} + \mathrm{e}^{\mathrm{i}\omega} u^{-}(v^{-})^{\mathsf{T}}\right). \tag{24}$$

Thus $\operatorname{rank} L \le 2$. We have

$$\|u\|_2^2 = \left\|u^{-}\right\|_2^2 = \frac{\ell_0^2}{2},$$

and the alternating odd-integer sum gives

$$\left|\left\langle u, u^{-}\right\rangle\right| = \frac{1}{2}\left|\sum_{j=0}^{\ell_0-1}(2j+1)(-1)^j\right| = \frac{\ell_0}{2}.$$

Because $1 \le a_k \le 2$, because $|\mathcal{K}| = \lfloor r/2 \rfloor$, and because every $k \in \mathcal{K}$ satisfies $k+1 \asymp r$,

$$c_1 r^2 \le \|v\|_2^2 = \left\|v^{-}\right\|_2^2 \le C_1 r^2.$$

Suppose first that $\ell_0 \ge 2$. Expanding the squared Hilbert–Schmidt norm in (24) and using Cauchy–Schwarz on its cross term gives

$$\begin{aligned}
\|L\|_{\mathrm{HS}}^2 &\ge \frac{1}{\omega^2}\left(2\|u\|_2^2\|v\|_2^2 - 2\left|\left\langle u, u^{-}\right\rangle\right|\|v\|_2\left\|v^{-}\right\|_2\right) \\
&= \frac{2\|v\|_2^2}{\omega^2}\left(\frac{\ell_0^2}{2} - \frac{\ell_0}{2}\right) \\
&\gtrsim \frac{\ell_0^2 r^2}{\omega^2}.
\end{aligned}$$

Because $L$ has rank at most two,

$$\|L\|_{\mathrm{op}} \ge \frac{\|L\|_{\mathrm{HS}}}{\sqrt{2}} \gtrsim \frac{\ell_0 r}{\omega}. \tag{25}$$

The triangle inequality, (23), and (25) yield

$$\begin{aligned}
\|A\|_{\mathrm{op}} &\ge \|L\|_{\mathrm{op}} - \|E\|_{\mathrm{op}} \\
&\ge c\frac{\ell_0 r}{\omega} - C\frac{\ell_0^2 r^2}{\omega^2} \\
&= \frac{\ell_0 r}{\omega}\left(c - C\frac{\ell_0 r}{\omega}\right).
\end{aligned}$$

Since $\ell_0 \le \ell$, taking the constant in $\omega \ge C_0 \ell r$ sufficiently large makes the parenthesis at least $c/2$. Thus

$$\|A\|_{\mathrm{op}} \gtrsim \frac{\ell_0 r}{\omega}.$$

By (22), this is $\gtrsim \ell r/\omega$.

It remains to handle $\ell_0 = 1$. Either $\ell = 1$, or $\lfloor r/2 \rfloor = 1$ and hence $r \le 3$. Since $\ell \le r$, in both cases

$$\ell \le 3. \tag{26}$$

For this case use the full row indexed by $j = 0$ rather than restricting the columns to $\mathcal{K}$. Define

$$B_{0k} := \int_{-1}^{1} \phi_0(t)\phi_k(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt, \qquad 0 \le k < r.$$

This is also a submatrix of $\Pi_\ell M_\omega \Pi_r$. Since $\phi_0 = 1/\sqrt{2}$,

$$\|B\|_{\mathrm{op}}^2 = \sum_{k=0}^{r-1} \frac{2k+1}{4} \left|F_k(\omega)\right|^2 .$$

Write

$$\varphi_k := \omega - \theta_k.$$

By Lemma 5.4, the principal term of $F_k$ is

$$\frac{2a_k}{\omega} \cos \varphi_k \quad \text{if } k \text{ is even,}$$

and

$$-\frac{2\mathrm{i}a_k}{\omega} \sin \varphi_k \quad \text{if } k \text{ is odd.}$$

For consecutive indices $0 \le k < r-1$,

$$|\varphi_{k+1} - \varphi_k| = |\theta_{k+1} - \theta_k| \le C\frac{2k+2}{\omega} \le C\frac{r}{\omega}.$$

After increasing the absolute constant in $\omega \ge C\ell r$, this increment is at most $1/10$. For every adjacent pair $(k, k+1)$ with $k$ even, the elementary inequality

$$\cos^2 s + \sin^2(s+\delta) \ge 1 - |\sin \delta|$$

shows that the sum of the squared principal amplitudes in the pair is at least $c/\omega^2$, because $a_k, a_{k+1} \ge 1$. The smaller of the two weights in that pair is $(2k+1)/4$. Summing over the disjoint pairs $(0,1), (2,3), \ldots$ gives

$$\begin{aligned}\sum_{k=0}^{r-1} \frac{2k+1}{4} \left|\text{principal term of } F_k\right|^2 &\gtrsim \frac{1}{\omega^2} \sum_{\substack{0 \le k < r-1 \\ k \text{ even}}} (2k+1) \\ &\gtrsim \frac{r^2}{\omega^2}.\end{aligned}$$

To include the remainders, use $|z+w|^2 \ge |z|^2/2 - |w|^2$. Since $|\rho_k| \lesssim \omega^{-2}$ and $\sum_{k<r}(2k+1) = r^2$,

$$\sum_{k=0}^{r-1} \frac{2k+1}{4} |\rho_k|^2 \lesssim \frac{r^2}{\omega^4}.$$

This is smaller than the principal lower bound after increasing the absolute constant in the hypothesis. Consequently,

$$\|B\|_{\mathrm{op}}^2 \gtrsim \frac{r^2}{\omega^2}.$$

Thus

$$\|B\|_{\mathrm{op}} \gtrsim \frac{r}{\omega} \gtrsim \frac{\ell r}{\omega},$$

where the last inequality uses (26).

In every case,

$$\|\Pi_\ell M_\omega \Pi_r\|_{2\to 2} \gtrsim \frac{\ell r}{\omega}.$$

To pass from the matrix norm to the stated integral, let unit vectors $a$ and $b$ satisfy

$$|a^* A b| = \|A\|_{\mathrm{op}}$$

in the case $\ell_0 \geq 2$; use $B$ in the one-row case. Set

$$p(t) := \sum_j \overline{a_j}\phi_j(t), \qquad q(t) := \sum_k \overline{b_k}\phi_k(t),$$

where coefficients outside the selected rows or columns are set to zero. Orthonormality gives $\|p\|_2 = \|q\|_2 = 1$, and

$$\begin{aligned}\int_{-1}^{1} p(t)\overline{q(t)}\mathrm{e}^{-\mathrm{i}\omega t}\,dt &= \sum_{j,k} \overline{a_j} b_k \int_{-1}^{1} \phi_j(t)\phi_k(t)\mathrm{e}^{-\mathrm{i}\omega t}\,dt \\ &= a^* A b.\end{aligned}$$

This proves the proposition. □

# 6 Proof of the main theorem

*Proof of Theorem 2.1.* If $\ell = r = 1$, choose $D \in [d, d+1]$ with fractional part $1/4$ and take

$$x(t) = 1, \qquad y(t) = \mathrm{e}^{2\pi\mathrm{i}Dt}.$$

Then the frequency separation is at least $d$ and

$$\begin{aligned}\left|\langle x, y\rangle_{[-1,1]}\right| &= \frac{|\sin(2\pi D)|}{\pi D} = \frac{1}{\pi D}, \\ \|x\|_2 \, \|y\|_2 &= 2.\end{aligned}$$

Therefore

$$\frac{\left|\langle x, y\rangle_{[-1,1]}\right|}{\|x\|_2 \, \|y\|_2} = \frac{1}{2\pi D} \gtrsim \frac{1}{d}.$$

Hence suppose $r \geq 2$. Choose

$$D := d + 1, \qquad \omega := 2\pi D.$$

After increasing the absolute constant in the hypothesis $d \geq C\ell r$, Proposition 5.8 gives unit polynomials $p \in \mathcal{P}_\ell$ and $q \in \mathcal{P}_r$ and an absolute constant $c_0 > 0$ such that

$$\left|\int_{-1}^{1} p(t)\overline{q(t)}\mathrm{e}^{-2\pi\mathrm{i}Dt}\,dt\right| \geq c_0 \frac{\ell r}{D}.$$

Apply Lemma 3.1 with

$$\eta \leq \min\left\{\frac{1}{2}, \frac{c_0 \ell r}{4D}\right\}.$$

The resulting signals have separation at least $d$, and

$$\begin{aligned}\left|\langle x, y\rangle_{[-1,1]}\right| &\geq \frac{3c_0\ell r}{4D}, \\ \|x\|_2 \, \|y\|_2 &\leq (1+\eta)^2 \leq \frac{9}{4}.\end{aligned}$$

Because $D = d + 1 \lesssim d$, we conclude that

$$\frac{\left|\langle x, y\rangle_{[-1,1]}\right|}{\|x\|_2 \|y\|_2} \gtrsim \frac{\ell r}{d}.$$

□

# 7 Scaling to a general observation interval

**Corollary 7.1.** *Let $T > 0$, $1 \le \ell \le r$, and $dT \ge C\ell r$. Then there exist separated Fourier-sparse signals on $[-T, T]$ such that*

$$\operatorname{dist}\big(\operatorname{supp} \widehat{x}, \operatorname{supp} \widehat{y}\big) \ge d$$

*and*

$$\left|\langle x, y\rangle_{[-T,T]}\right| \ge c\frac{\ell r}{dT} \|x\|_{L^2[-T,T]} \|y\|_{L^2[-T,T]}.$$

*Proof.* Apply Theorem 2.1 on $[-1, 1]$ with separation parameter $dT$, and denote the resulting signals by $X$ and $Y$. Define, for $t \in [-T, T]$,

$$x(t) := X(t/T), \qquad y(t) := Y(t/T).$$

If $f$ is a frequency of $X$, then $f/T$ is the corresponding frequency of $x$; the same holds for $Y$ and $y$. Hence

$$\operatorname{dist}(\operatorname{supp} \widehat{x}, \operatorname{supp} \widehat{y}) = \frac{1}{T} \operatorname{dist}(\operatorname{supp} \widehat{X}, \operatorname{supp} \widehat{Y}) \ge d.$$

With the change of variables $s = t/T$,

$$\begin{aligned}
\langle x, y\rangle_{[-T,T]} &= T \langle X, Y\rangle_{[-1,1]}, \\
\|x\|_{L^2[-T,T]} &= \sqrt{T} \|X\|_{L^2[-1,1]}, \\
\|y\|_{L^2[-T,T]} &= \sqrt{T} \|Y\|_{L^2[-1,1]}.
\end{aligned}$$

Substitution into Theorem 2.1 gives the claimed inequality. □

# 8 Why the elementary proof loses the unbalanced regime

The endpoint-kernel proof estimates

$$\big\|(pq)''\big\|_1 \lesssim \ell r(\ell^2 + r^2),$$

so two integrations by parts require modulation at least of order $\ell^2 + r^2$. When $r \gg \ell$, this is much larger than the sharp scale $\ell r$.

The Bessel-phase proof uses more structure. In the positive Legendre linearization of $P_j P_k$, only orders

$$k - j \le n \le k + j$$

appear. Across this short window, the residual Debye phase varies by only

$$O\left(\frac{jk}{\omega}\right).$$

Thus the entire product transform remains coherent already at

$$\omega \gtrsim \ell r.$$

After normalization, the leading cross-Gram matrix is the sum of two rank-one matrices of operator norm $\Theta(\ell r/\omega)$; the remaining error is smaller by another factor $\ell r/\omega$. This is the mechanism that survives in the highly unbalanced case.

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